

JAVA PRIME PALINDROME PROGRAM

Object-Oriented Programming Practical Record

1. Aim

To write a Java program using command line arguments to check whether a given number is Prime, Palindrome, or both.

2. Steps for Execution

1. Open VS Code, Eclipse or IntelliJ IDEA with JDK installed.
2. Create a Java file named PrimePalindrome.java.
3. Enter the Java source code and save the file.
4. Compile the program using `javac PrimePalindrome.java`.
5. Run the program using `java PrimePalindrome 131`.
6. The program checks whether the number is Prime.
7. The program checks whether the number is Palindrome.
8. Display the final result.

3. Algorithm

1. Start the program.
2. Read the number from command line argument.
3. Check whether the number is Prime.
4. Store the original number.
5. Reverse the number using a loop.
6. Compare the original number with the reversed number.
7. If both conditions are true, print Prime and Palindrome.
8. Otherwise print the appropriate result.
9. Stop the program.

4. Explanation of the Program

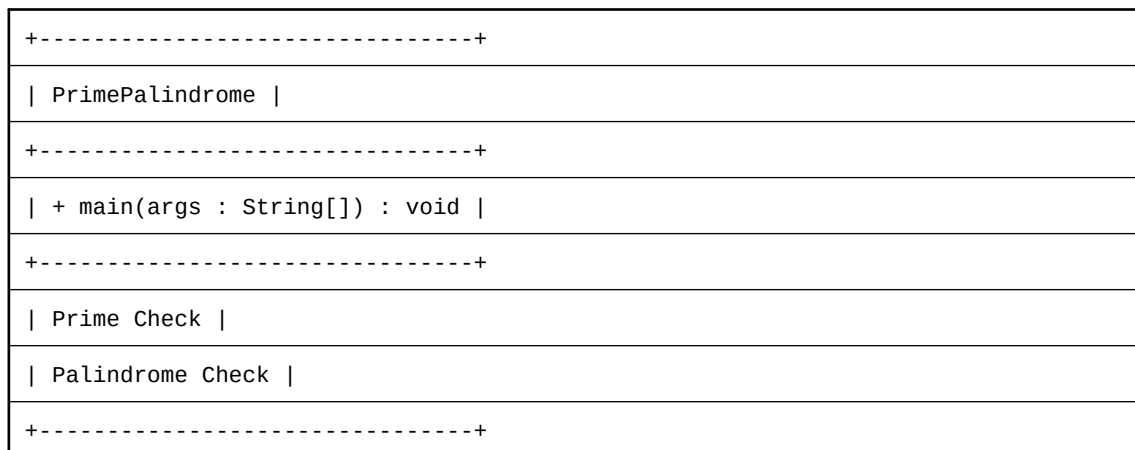
Command Line Argument: The number is received through args[0].

Prime Check: The program checks whether the number is divisible by any number from 2 to n/2.

Palindrome Check: The digits are reversed using a while loop and compared with the original number.

Result: The program displays whether the number is Prime, Palindrome, both, or neither.

5. UML Class Diagram



6. Java Source Code

```
public class PrimePalindrome {  
  
    public static void main(String[] args) {  
  
        if (args.length == 0) {  
            System.out.println("Please enter a number as command line argument.");  
            return;  
        }  
  
        int n = Integer.parseInt(args[0]);  
  
        boolean prime = true;  
  
        if (n <= 1) {  
            prime = false;  
        } else {  
            for (int i = 2; i <= n / 2; i++) {  
                if (n % i == 0) {  
                    prime = false;  
                    break;  
                }  
            }  
        }  
  
        int original = n;  
        int reverse = 0;  
        int temp = n;  
  
        while (temp > 0) {  
            int digit = temp % 10;  
            reverse = reverse * 10 + digit;  
            temp = temp / 10;  
        }  
    }  
}
```

```
boolean palindrome = (original == reverse);

System.out.println("Number = " + original);

if (prime && palindrome)
    System.out.println(original + " is both Prime and Palindrome.");
else if (prime)
    System.out.println(original + " is Prime but not Palindrome.");
else if (palindrome)
    System.out.println(original + " is Palindrome but not Prime.");
else
    System.out.println(original + " is neither Prime nor Palindrome.");
}
```

7. Output of the Program

Example 1: Input = 131

```
Number = 131  
131 is both Prime and Palindrome.
```

Example 2: Input = 25

```
Number = 25  
25 is Palindrome but not Prime.
```

Example 3: Input = 20

```
Number = 20  
20 is neither Prime nor Palindrome.
```

8. Result

Thus, the Java program successfully checks whether the given number is Prime and Palindrome using command line arguments.